



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

15 September 2017

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the question paper.

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
1	/ 12
2	/ 9
3	/ 9
4	/ 14
5	/ 10
6	/ 14
7	/ 12
8	/ 9
9	/ 11
Total	/ 100

This paper consists of **23** printed pages.

[Turn over]

QUESTION 1

- (a) In 1934, two biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.

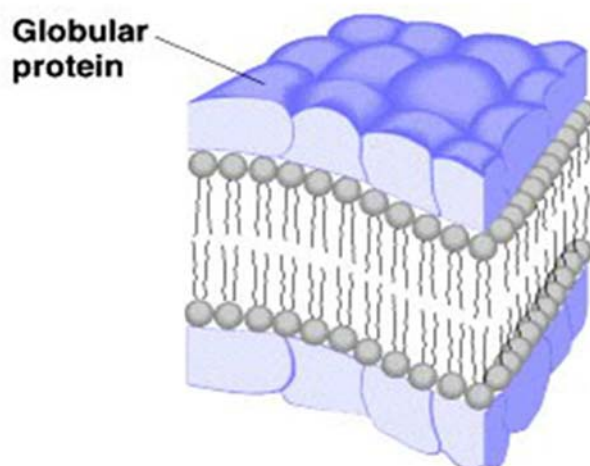


Fig. 1.1

- (i) State one way in which the Davson-Danielli structure is similar to the fluid mosaic structure and one way in which it differs from the fluid mosaic model. [2]

[Similarity – max 1]

- Phospholipids are arranged in a bilayer
- Both contain peripheral proteins on both sides of the membrane

[Difference – max 1]

- The proteins in the Davson-Danielli structure are peripheral / outside the bilayer while the proteins in the fluid mosaic model are both integral and peripheral to the bilayer
- **Idea that** Proteins in the Davson-Danielli structure are in **uniform locations** across the surface of the membrane but proteins in the fluid mosaic model are found in **different locations and numbers** across the membrane
- In the fluid mosaic model, the membrane contains other molecules like cholesterol but the membrane in the Davson-Danielli structure does not.
- In the fluid mosaic model, the membrane contains glycoproteins / glycolipids but the membrane in the Davson-Danielli structure does not.

- (ii) Suggest two problems that the Davson-Danielli structure of the membrane would pose to the functioning of the cell. [2]

- Limited regulation of membrane fluidity in the absence of cholesterol
- Not able to transport hydrophilic substances across the membrane as there are no transmembrane proteins present
- No transmembrane receptors to transmit signals from the extracellular environment
- No glycoproteins for cell-to-cell recognition / adhesion
- AVP

(b) Transport of substances across membranes involve many different mechanisms.

Fig. 1.2 is a diagram showing the transport of protein-rich solid particles into an animal cell.

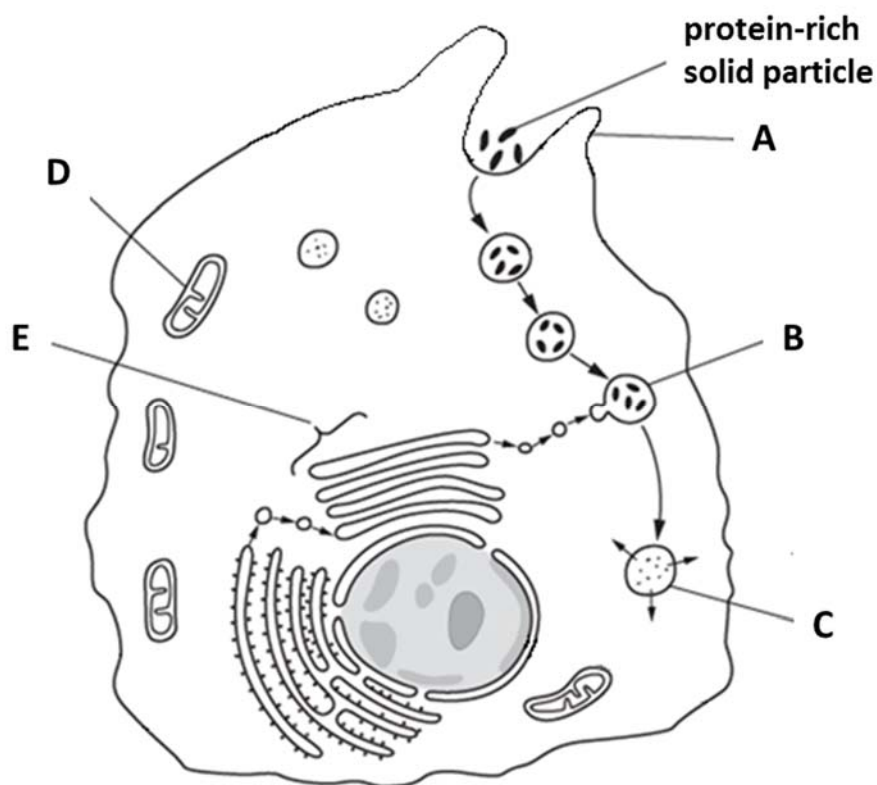


Fig. 1.2

(i) Describe the process at A. [2]

- Phagocytosis - the cell surface membrane extends outwards to form pseudopodia that surrounds the solid particle
- The pseudopodia then fuse to form a phagosome / phagocytic vesicle around the particle

(ii) Describe what happens to the protein-rich solid particle between B and C. [2]

- The primary lysosome fuses with the phagosome to form a secondary lysosome
- The protein-rich solid particles are hydrolysed by lysosomal enzymes/proteases into many amino acids that are transported / released out of C to the cytosol

(iii) Name the organelles **D** and **E** and briefly describe their roles in the formation of **C**. [4]

[Organelle **D**]

- The mitochondria
- Produces ATP is needed as an energy source for the movement of (primary) lysosomes and phagosomes so that they can fuse

[Organelle **E**]

- The Golgi apparatus
- is the site of biochemical modification e.g. phosphorylation / cleavage of lysosomal enzymes to make them functional
- It also sorts and packages the lysosomal enzymes at the trans face into primary lysosomes

[Total: 12]

QUESTION 2

Glucose is phosphorylated at the start of glycolysis by the tetrameric enzyme, hexokinase.

There are multiple hexokinase isozymes (I-IV) for the phosphorylation of glucose, enabling specific organs to regulate carbohydrate metabolism in a unique way. Hexokinase IV, also called glucokinase, is the predominant isozyme in the liver, while hexokinase I is found in almost all other tissues.

Fig. 2.1 shows the difference in fractional saturation between glucokinase and hexokinase I, which represents the fraction of binding sites that are occupied by glucose.

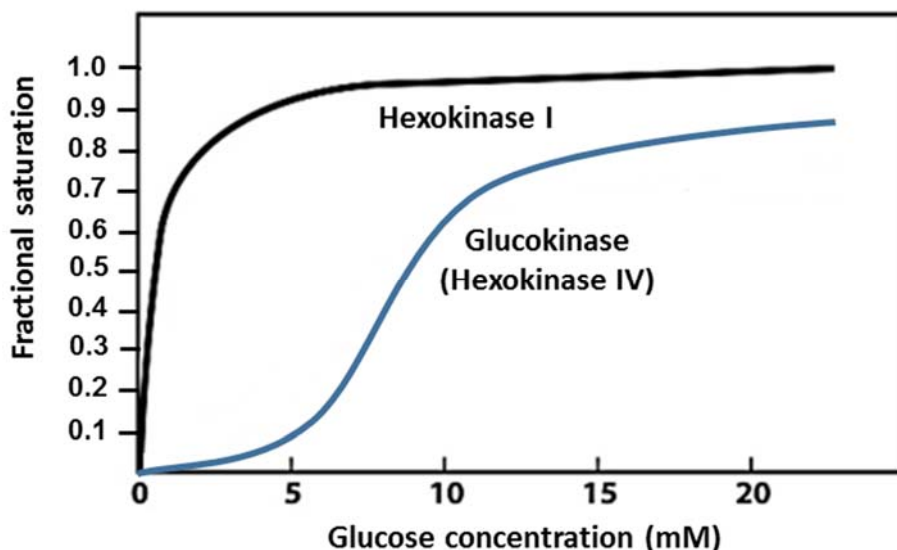


Fig. 2.1

(a) With reference to Fig. 2.1 and using your knowledge of enzymes, account for the shape of the curve for glucokinase from 0 to 10mM of glucose concentration. [4]

- **[data citation]** From 0-5mM glucose concentration, the fractional saturation of glucokinase increases gradually from 0 to 0.1 then from 5 to 10mM glucose concentration, fractional saturation increases sharply from 0.1 to 0.65 (accept 0.6 to 0.66)
- This is because glucokinase is an allosteric enzyme comprising four subunits / tetrameric
- Glucokinase has low affinity for glucose / usually in its inactive form
- Cooperative binding of glucose: As glucose concentration increases from 5 to 10mM, the binding of glucose to the active site of one subunit causes a conformational change to all the other subunits to their active forms

(b) Suggest an advantage to most cells in the body of containing hexokinase I rather than glucokinase. [2]

- **Ref. to** high affinity of hexokinase I for glucose
- This allows hexokinase I to phosphorylate glucose even when glucose concentrations in the cell are low.

(c) During a sporting event, muscle cells of an athlete may have to carry out respiration in anaerobic as well as aerobic conditions to produce sufficient ATP.

(i) Name the membrane-bound enzyme responsible for producing ATP from ADP and inorganic phosphate. [1]

- ATP synthase

(ii) Explain how anaerobic respiration helps to meet the demand for sufficient ATP. [2]

- Pyruvate is converted to lactate by lactate dehydrogenase to regenerate NAD⁺
- This allows glycolysis to continue to produce 2 net ATP via substrate-level phosphorylation.

[Total: 9]

QUESTION 3

In vertebrates, sister chromatid cohesion is dependent on a complex of proteins called cohesin, which binds to and joins sister chromatids at the centromere until the onset of anaphase.

Sister chromatid separation is initiated by cleavage of cohesin by the enzyme separase. Prior to anaphase, a protein called securin binds to separase and maintains it in the inactive form. Anaphase is initiated when securin is degraded, freeing the enzyme separase.

Fig. 3.1 illustrates the transition from metaphase to anaphase during a mitotic cell cycle.

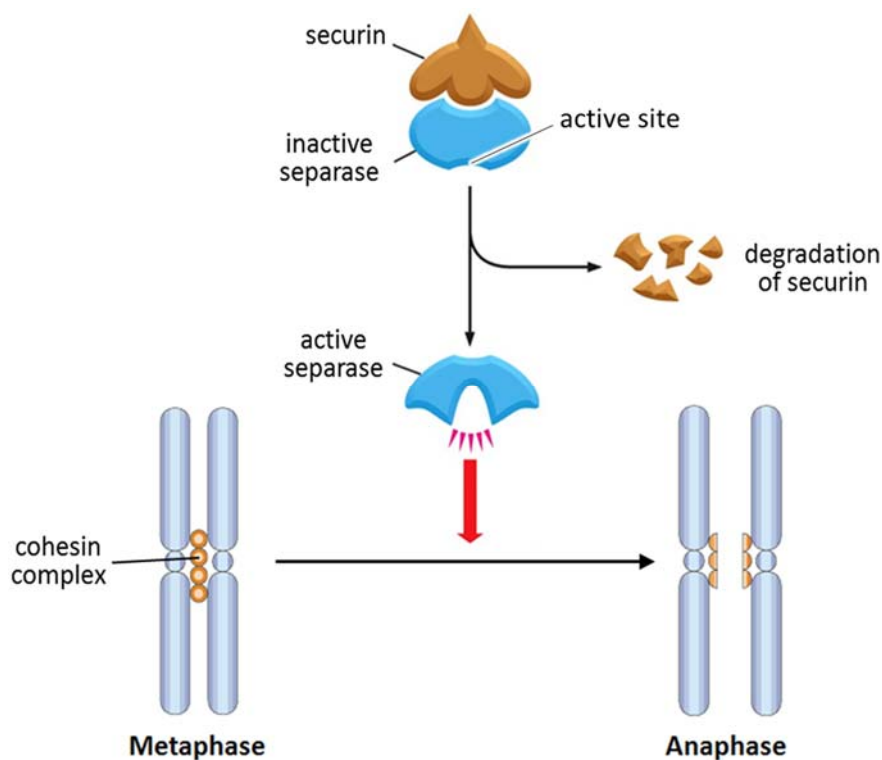


Fig. 3.1

(a) State a feature of centromeric DNA. [1]

- Non-coding repetitive sequence of DNA

(b) Explain how a mutation to the centromeric DNA can lead to aneuploidy. [3]

- Change in sequence of centromeric DNA
- Shape of mutated centromeric sequence is no longer complementary to kinetochore protein complex, hence they cannot bind,
- Microtubules / spindle fibres are unable to attach to the centromere via the kinetochore
- Nondisjunction occurs, where sister chromatids are not separated properly, giving rise to cells with extra or missing chromosome(s) / $2n+1$ and $2n-1$.

OR

- Change in sequence of centromeric DNA
- Shape of mutated centromeric sequence is no longer complementary to cohesin proteins, hence cohesin cannot bind.
- Sister chromatids are not held together prior to anaphase and so spindle fibres from one pole may attach to both sister chromatids.
- Nondisjunction occurs when two sister chromatids are pulled to one pole, giving rise to cells with extra or missing chromosome(s) / $2n+1$ and $2n-1$.

(c) State the class of enzyme to which separase belongs. [1]

- Proteases

(d) Explain how securin maintains separase in the inactive form. [3]

- Allosteric / non-competitive inhibitor which binds to a site away from the active site / allosteric site
- Changes conformation of separase and hence the shape of its active site
- Active site no longer complementary in shape to its substrate cohesin and hence cannot bind

(e) Explain how securin is degraded at the onset of anaphase. [1]

- Securin is tagged with ubiquitin proteins / are ubiquitinated and sent to the proteasome for degradation.

[Total: 9]

QUESTION 4

Takahashi and fellow scientists had successfully reprogrammed human fibroblasts into a pluripotent state, known as induced pluripotent stem cells (iPS cells).

(a) Define the term *pluripotent stem cells*. [1]

- Have the ability to differentiate into almost any cell type to form any organ / except cells of the extraembryonic membranes.

(b) The generation of iPS cells made use of four protein factors (non-enzymatic proteins), which are introduced into differentiated cells by retroviruses. Research has proven their function in upregulating “stemness” genes, while suppressing differentiation-associated genes in human iPS cells.

(i) State the general name given to proteins such as those four protein factors. [1]

- Specific transcription factors

(ii) Explain why the protein factor involved in upregulating “stemness” genes will contain both a DNA-binding domain and a protein-binding domain. [2]

- DNA-binding domain binds to specific DNA/nucleotide sequences that makes up the enhancer.
- Protein-binding domain binds to and stabilizes the transcription initiation complex, to increase the rate of transcription.

(iii) Explain how an amino acid substitution within the DNA-binding domain can affect the function of the protein factor in (b)(ii). [3]

- The amino acid substituted has a different R group with different properties.
- This will affect R group interactions / bonds between the amino acids that maintains the 3D conformation / structure of the protein.
- Tertiary structure/ 3D conformation of DNA-binding domain no longer complementary to the shape of specific DNA sequences/ nucleotides of enhancer.

(iv) Explain why it is important that the scientists ensure high telomerase activities in the iPS cells. [3]

- (Induced) pluripotent stem cells need to undergo continuous cell division / cell renewal.
- **Idea that** Telomerase must be present to elongate the telomeres which are shortened after every round of (DNA) replication. [R: “Prevent end-replication problem”]
- This allows telomere length to be maintained / prevents the loss of genes through erosion at chromosomal ends / prevents telomeres from reaching critical length, preventing apoptosis.

(c) Further studies had shown that some iPS cells developed tumors, which is often attributed to the use of retrovirus. This issue of tumourgenesis must be overcome before iPS cells can be used in human therapies.

(i) Suggest why the development of tumor in the iPS cells may be attributed to the use of retrovirus. [2]

- Retroviruses are capable of random integration of (viral) DNA into the host genome, disrupting proto-oncogenes / tumour suppressor genes.
- Result in the gain of function mutation of proto-oncogene / loss of function mutation of tumor suppressor gene. Resulting in uncontrolled cell division, forming tumor.

(ii) With reference to the benefits and problems of iPS cells, discuss whether research on iPS cells should be continued. [2]

Mark for ideas

No [1]

- Higher risk of tumor formation due to the use of retroviruses.
- Other logical reason.

Yes [1]

iPS cells possess development potential of the embryonic stem cells....

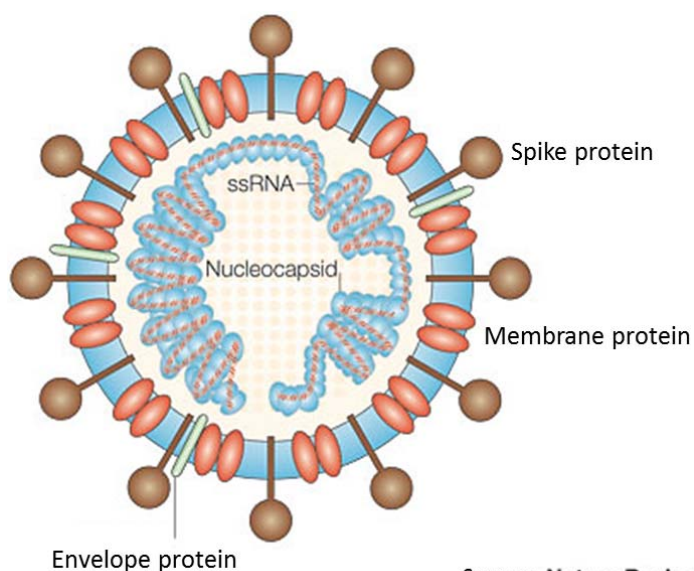
- Can be differentiated into all cell types (except embryonic membrane) and use for stem cell therapy.
- Can also be used to differentiate into disease-specific cell type for drug / toxicity testing on the effectiveness and risks involved when using the drug.
- Able to differentiate into patient-specific cell type for drug / toxicity testing to test the effectiveness and possible reaction of patient's cells to the drugs.
- Use of iPS cells avoid ethical issues as embryos are not used / no destruction of embryos.
- Tissue rejection can also be avoided/lower chances of immune response, since the iPS cells can be induced from differentiated cells of the patient (patient-specific cell type).

[Total: 14]

QUESTION 5

In May 2014, the Middle East respiratory syndrome coronavirus (MERS-CoV), which was first reported in Saudi Arabia in 2012, infected two Americans who travelled to Saudi Arabia.

Coronaviruses are enveloped RNA viruses that infect and cause lower respiratory tract disease in a broad array of animals and humans. Virus particles range from 70 to 120 nm in diameter and are surrounded by characteristic spike-shaped glycoproteins, as shown in Fig. 5.1. Coronaviruses contain the largest single-stranded, positive-strand RNA genomes currently known, which range from 25.5 to nearly 32 kb in length.



Source: **Nature Reviews | Immunology**

Fig. 5.1

(a) Describe **two** structural differences between the genome of the coronavirus and the influenza virus. [2]

- Genome in influenza virus is negative-strand RNA, while that in coronavirus is positive-strand RNA.
- Eight, separate single-stranded RNA in influenza virus, while there is only one continuous long RNA strand in coronavirus

(b) Describe how the coronavirus enters its host cell.

[3]

[Entry by fusion]

- Spike protein / Glycoprotein is complementary in shape to certain cell surface receptor on the host cell.
[Reject if the idea that the receptor is on the host cell is unclear/not present]
- Binding triggers conformational change to the viral envelope protein which in turn result in the fusion of the viral envelope with the host cell surface membrane is triggered.
- The nucleocapsid/RNA genome released into the cytosol.

OR

[Entry by RME]

- Spike protein / Glycoprotein is complementary in shape to certain cell surface receptor on the host cell.
[Reject if the idea that the receptor is on the host cell is unclear/not present]
- Virus then enters the host cell via receptor-mediated endocytosis, where the host cell membrane forms an endosome/endocytotic vesicle around the virus.
- Fusion of the viral envelope with the endosomal membrane releases the nucleocapsid/RNA genome into the cytosol.

(c) Describe the process which allows the coronavirus to infect a *broad array of animals and humans* overtime. [2]

- Antigenic drift occurs: Gene coding for spike protein/glycoprotein undergoes mutation.
- Changes in conformation to the spike protein which can bind to various receptors of different cell types / species.

(d) Unlike the human immunodeficiency virus, the coronavirus genome is not integrated into its host DNA.

Suggest how the coronavirus produces more copies of its genome.

[2]

- (+)RNA acts as a template to produce (-)RNA, which in turn acts as a template to produce many copies of the (+)RNA genome...
- ...by viral RNA-dependent RNA polymerase.

[catalysed by replicase, accept if student mention viral RNA-dependent RNA polymerase]

(e) The fatality rate of coronavirus infections is approximately 60%.

Briefly explain how the coronavirus can cause death in humans.

[1]

- Since CoV infects and damages epithelial cells of the lower respiratory tract → Suffocation / respiratory failure leading to death.

[Total: 10]

QUESTION 6

- (a) Fig. 6.1 is an electron micrograph of a process that bacterial cells undergo which results in the formation of two daughter cells.

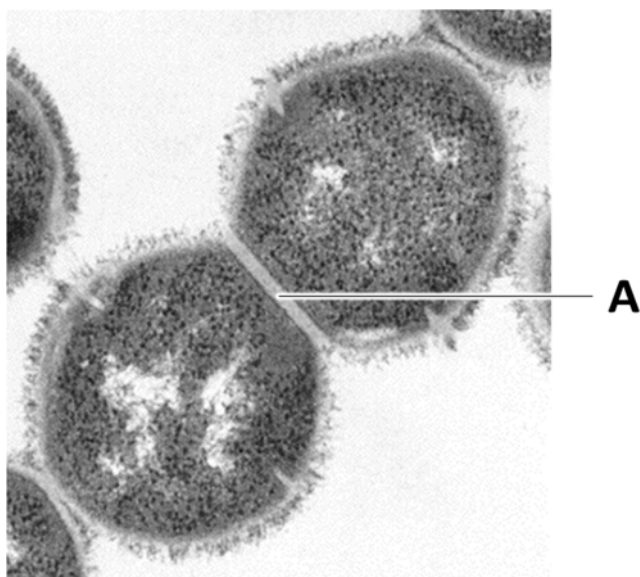


Fig. 6.1

- (i) Name the process above and state the main component making up structure A. [1]

Process: Binary fission

Component making up Structure A: Peptidoglycan cell wall

[Both correct – 1m]

- (ii) “The process above will always produce two genetically identical daughter cells”.

Comment on the validity of this statement. [1]

[Not always true]

- No mechanism for equal division of plasmids during binary fission, hence plasmids may not be equally divided between daughter cells.
- No formation of spindle fibers in prokaryotes to pull chromosomes to opposite poles of the cell, hence may result in unequal division of chromosome.

[True]

- **Idea that** Bacterial chromosome has a point of attachment to the bacterial membrane which ensures equal separation of bacterial chromosomes during binary fission.

- (b) The *xyl* operon is a catabolic operon involved in the breakdown of the sugar xylose. Fig. 6.2 shows how a *xyl-lac* fusion operon is constructed, which consist of 2 structural genes from *lac* operon, regulatory sequences and the regulatory gene of the xylose operon. The arrows indicate the direction of transcription.

To test its effects, the fusion operon was constructed and packaged into bacteriophages. The fusion operon was then inserted into the chromosomes of these bacterial cells upon infection.

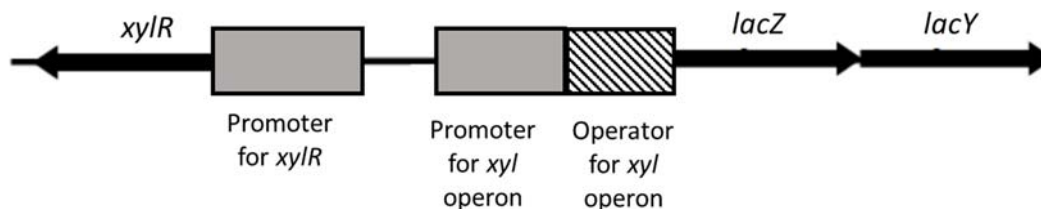


Fig. 6.2

- (i) State the process of this gene transfer. [1]

- Specialised Transduction

- (ii) Suggest and explain one advantage of the process stated in b(i) over transformation in bacteria. [2]

- In transduction, the operon can be integrated into the chromosome of bacteria but it may not be integrated in transformation.
- **Idea that** Hence operon replicate when bacteria chromosomes replicate, hence all daughter bacterial cells possess the operon.
OR
- **Idea that** Hence not easily degraded, ensuring more stable expression.
- In transduction, operon is packaged into phage capsid but in transformation, operon remains as naked DNA.
- **Idea that** Hence phage capsid protects the operon from degradation (by nucleases outside the bacteria cells)
- The rate of successful gene transfer is higher in transduction than in transformation.
- **Idea that** Phages target bacterial cells specifically / by nature.

- (iii) Explain the condition required for *lacZ* gene to be expressed in bacteria cells in which the *xyl-lac* fusion operon has been introduced. [3]

- Presence of xylose
- **Xylose act as inducer, which bind to and inducing a conformation change to inactivate xyl repressor.
- Inactive repressor cannot bind to operator, allowing RNA polymerase to bind to promoter to transcribe the *lacZ* gene.

*Award max 1m (3rd point only) if student identified presence of lactose.

(iv) Suggest why the direction of transcription of the regulatory and structural genes may differ.

KU-3

[2]

- **Idea that** Different strands was used as template for transcription.
- Since the strands (used) are antiparallel, the two strands are read in 3'→5' direction in opposite orientations.

(c) Colibacillosis is a fatal condition caused by *E. coli* in poultry. In a study to examine the effectiveness of bacteriophages in treating colibacillosis, broiler chickens were first subjected to an aerosol spray containing bacteriophages *on day 0*. They were then separated into five treatment groups. Each treatment group was subsequently injected with *E. coli* on days 0, 1, 2, 3 and 4 respectively. The mortality rate for each treatment group was determined after 21 days. The result of the study is represented by Fig. 6.3 below.

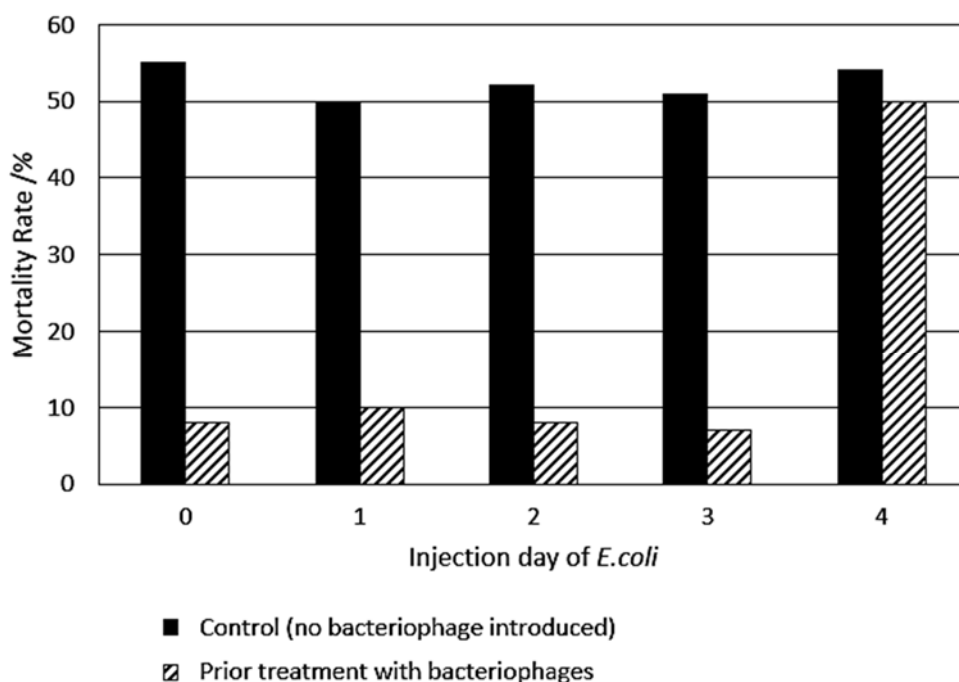


Fig. 6.3

With reference to Fig. 6.3 above,

(i) Compare the trends observed in the control group and the groups that have been treated with bacteriophages, and comment on the effectiveness of such treatment. [3]

[max 2]

- For injection days 0 – 3, the mortality rate for control group is consistently high, but low for treatment group.
- Mortality rate for control group ranges from 50-55% / averages 53 % (accept 52%) while mortality rate for treatment group ranges from 7(8)–10% / averages 8% (accept 9%).
- On Day 4, there is a spike in mortality rate for the treated chicken to 50% which is comparable to that of control which is 53%

(Accept and award pt 1 &/or 2 if students compared day 0-3 individually)

- **[compulsory]** Bacteriophages is effective in reducing mortality rate caused by colibacillosis for only 3 days before another dose is needed. [1]

(ii) Suggest why the use of bacteriophages is a better alternative to antibiotic therapy for the chickens. [1]

Idea that:

- Bacteriophage is more specific in targeting their host bacteria (complementary receptors), while antibiotics tend to have wider host range.
- Possible emergence of antibiotic resistant bacteria due to prolonged/inappropriate use of antibiotics, but no / low possibility of bacteriophage-resistant bacteria since they naturally infect bacteria.
- Bacteriophages are harmless to chicken/human, while antibiotic may trigger allergy response.

(Also applicable to humans since humans are consumer of the chickens.)

[Total: 14]

QUESTION 7

- (a) To analyse specific DNA sequences, various molecular processes are carried out. Fig. 7.1 below shows one possible way in which the gene for colour vision can be obtained for analysis.

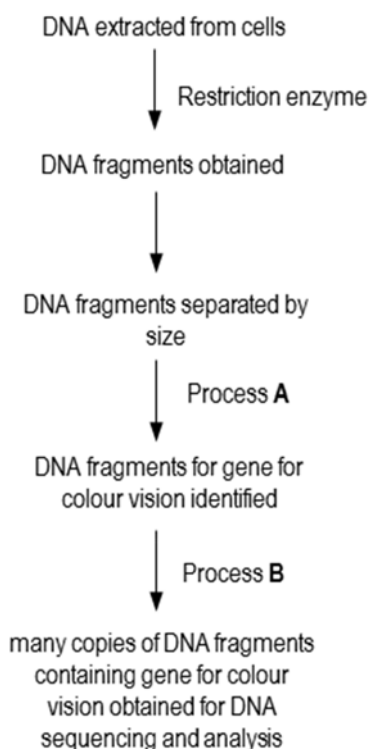


Fig. 7.1

- (i) Describe what is added to process **A** to identify the DNA fragment containing the gene for colour vision. [1]

- Radioactively-labelled / Fluorescently-labelled, single-stranded probe, complementary to the gene for colour vision.

- (ii) Explain how process **B** ensures that many copies of the target sequence is produced. [1]

- (DNA) primers added is complementary to the 3' region of the target sequence, flanking the gene for colour vision.

(b) The inheritance of colour vision and ABO blood group was analysed in an extended family.

The gene for colour vision is sex-linked.

The gene for the ABO blood group system is on chromosome 9. There are three alleles controlling blood group. These three alleles gives four possible phenotypes.

Fig. 7.2 shows the inheritance of these two genes in a family. Colour blindness is a rare condition, and can be assumed that the disease allele is not present in phenotypically normal individuals from other families.

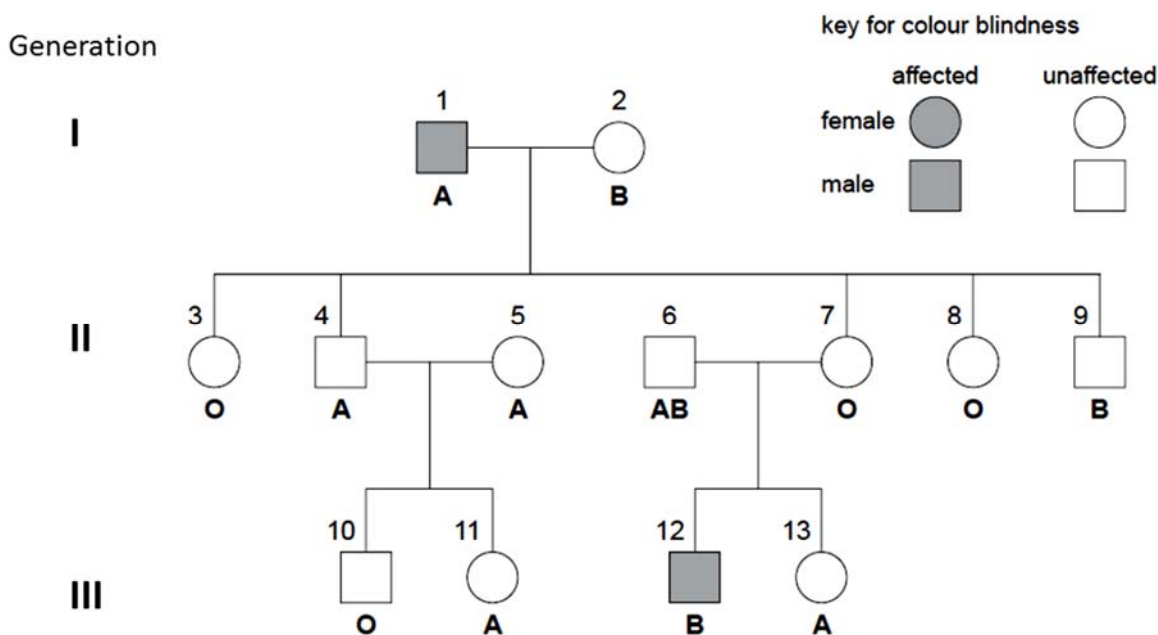


Fig 7.2

(i) State a possible genotype for each of the following people in the family shown in Fig. 7.2.

[2]

Individual I-2 $I^B i^O X^R X^R$ or $I^B i^O X^R X^r$

Individual II-9 $I^B i^O X^R Y$

[Reject: if symbols for blood groups are wrong / if colourblindness is not represented as sex-linkage]

[Reject: if symbol A, B and O is used to represent gene for colour vision]

(ii) With reference to Fig. 7.2, explain why one grandson (III-12) of individual I-1 has inherited colour blindness but the other (III-10) has not. [3]

- (Grandson) III-12 inherited the X^r allele / colour-blindness from his mother II-7, who is a carrier/heterozygote.
- She has inherited the X^r allele from her father, I-1.
- **Idea that** Grandson III-10 however is not able to inherit the X^r allele from the paternal line.

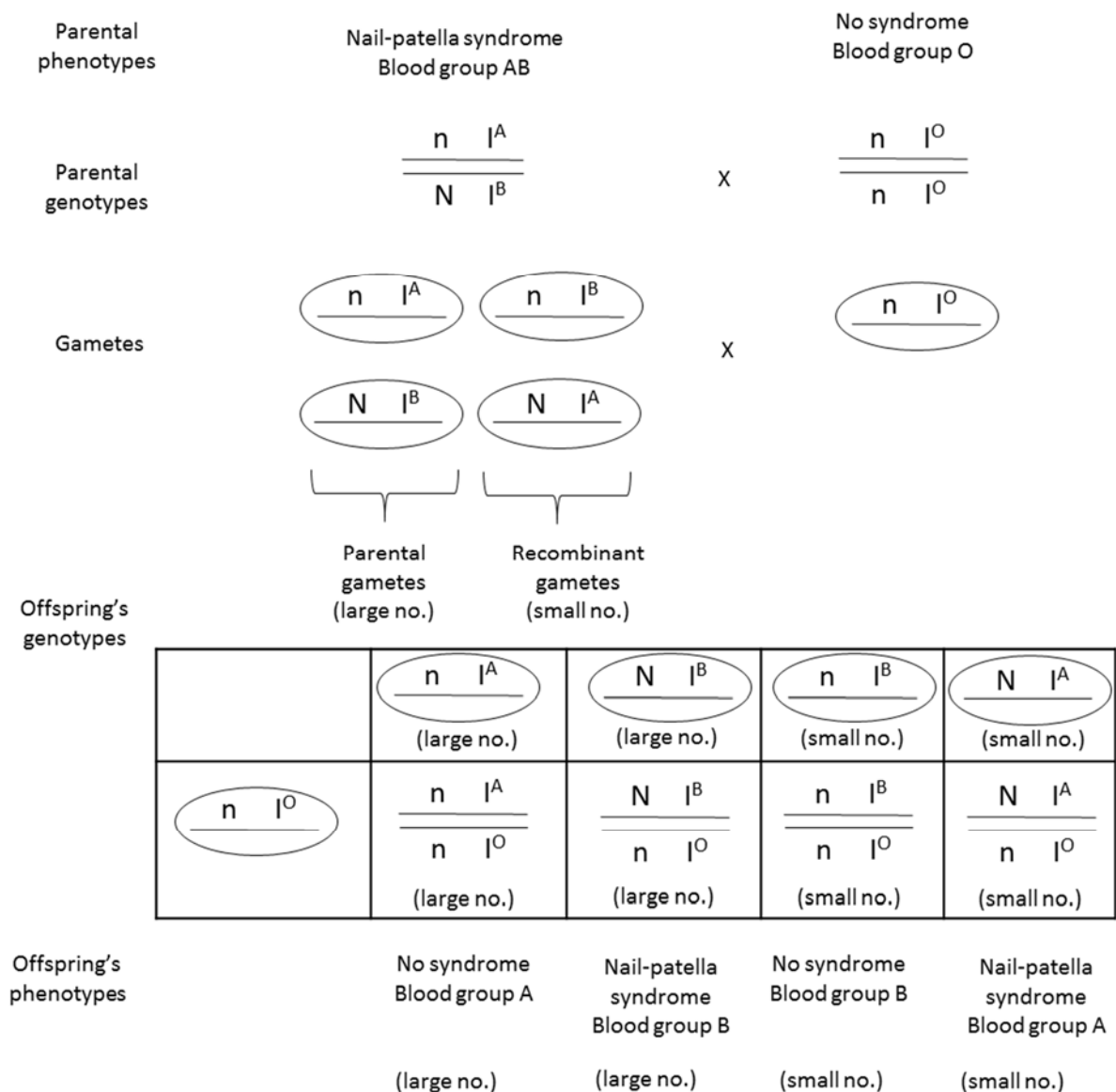
- (c) Nail-patella syndrome is a rare autosomal dominant trait that affects fingernails, toenails, elbows and kneecaps. The locus of the gene for nail-patella syndrome, **N** / **n**, is 10 map units from the ABO locus on chromosome 9.

A man with nail-patella syndrome and blood group AB has a family of five children with his wife who does have the syndrome and is blood group O.

Three children do not have the nail-patella syndrome and are blood group A.

Two children have nail-patella syndrome and are blood group B.

- (i) Illustrate the above cross between the man and his wife with a genetic diagram. [3]



- Correct parent genotypes
- Gametes (with labeling of parental (large no) & recombinant (small no).
- Offspring's genotypes and phenotypes (accept w/o indicating large/small no)

(*penalize 1m if symbols used is incorrect – for either or both genes)

(ii) Suggest why there is only a 5% probability of these parents having a child with both blood group A and nail-patella syndrome. [2]

- In the father, I^A allele and n allele are linked / I^B allele and N allele are linked [Reject: just saying that the two loci are linked]
- [mechanism] Since the two loci are 10 map units apart, the recombination frequency between the two loci is only 10% / 10% chance of crossing over.
- [outcome] Since there are two recombinant phenotypes, each phenotype comprises half of the 10% recombinant phenotypes.

[Total: 12]

QUESTION 8

Fig. 8.1 shows how a rise in blood glucose concentration stimulates the beta cells in the pancreas to secrete insulin, a protein hormone.

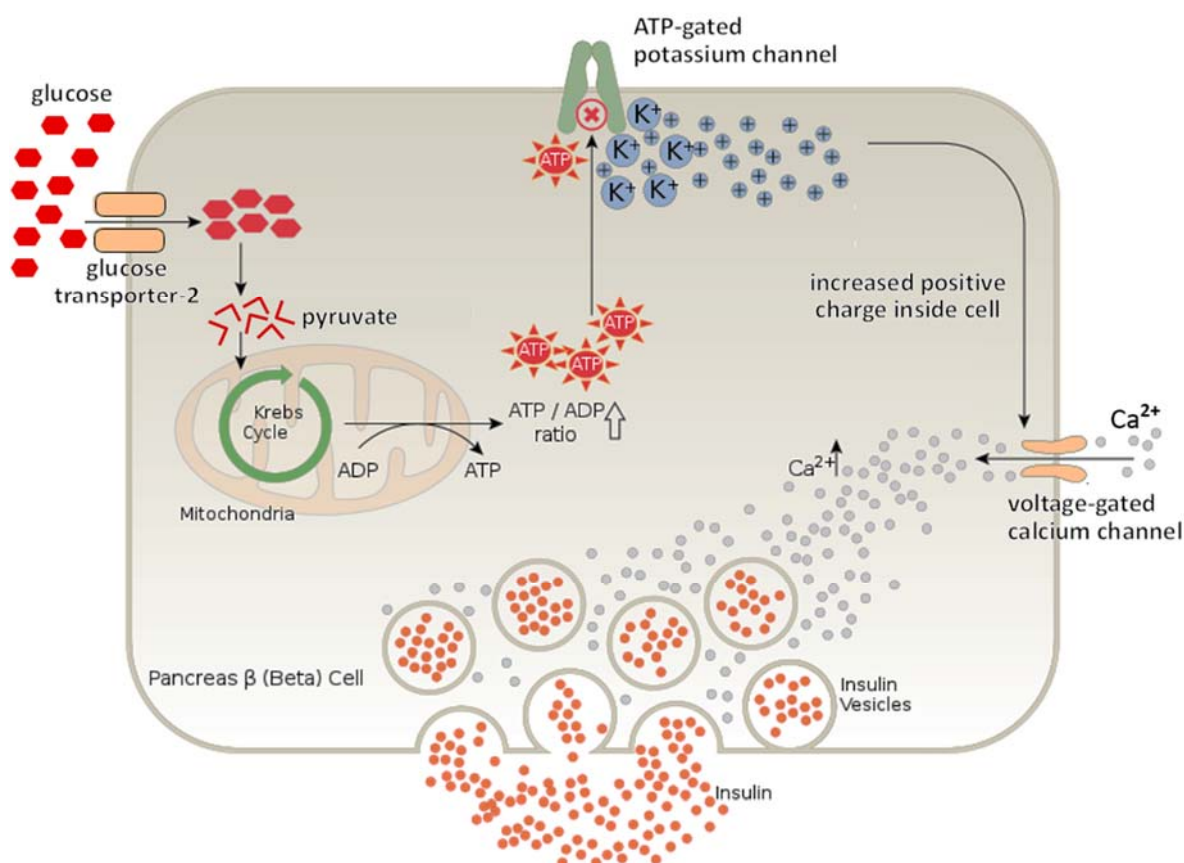


Fig. 8.1

(a) Explain the significance of an existing pool of insulin-rich vesicles in the β-cell. [2]

- Time is needed for insulin gene to be expressed and the insulin to be packaged into secretory vesicles
- β-cell need to respond very quickly to the sudden rise in blood glucose

(b) Outline the events leading to the release of insulin. [4]

- Glucose taken into the β-cell via glucose transporter 4 by facilitated diffusion [mark once]
- Glucose undergo aerobic respiration to produce ATP, which is transported out of mitochondria into cytosol to increase in cytosolic ATP / increase ATP:ADP ratio.
- ATP binds to ATP-gated potassium channel, causing it to close, leading to an accumulation of positive charge inside the cell
- This increased voltage causes voltage-gated calcium channel to open, and calcium ions enter the cell by facilitated diffusion [mark once]
- An increased in cytosolic calcium ions stimulate the insulin-rich vesicles to undergo exocytosis to release insulin.

Insulin released by β -cells reaches their target cells, such as liver and muscle cells. One of the responses of insulin is glycogen synthesis, as shown in Fig. 8.2.

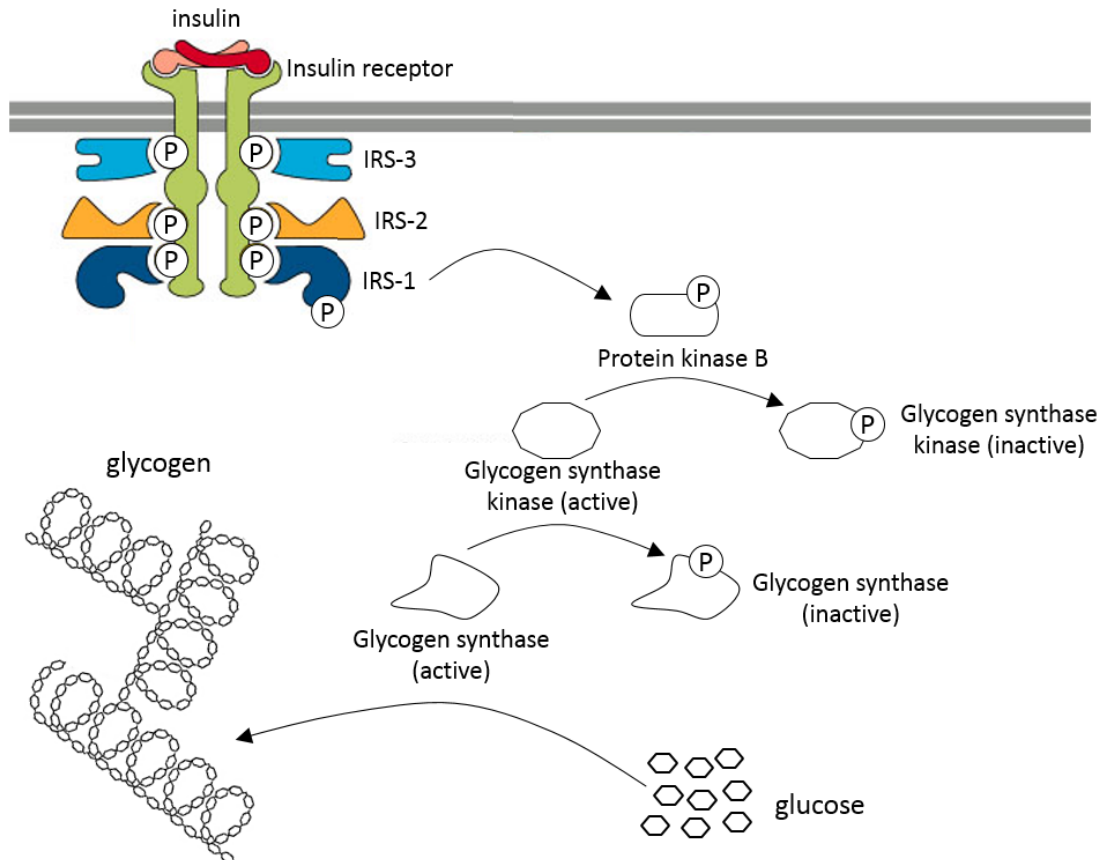


Fig. 8.2

(c) Describe how protein kinase B triggers glycogen synthesis. [3]

- PKB phosphorylates (active) glycogen synthase kinase (GSK), changing its conformation and inactivating it.
- Inactive GSK unable to phosphorylate and inactivate glycogen synthase (GS).
- GS remains active and hence convert polymerizes glucose to form glycogen.

[Total: 9]

QUESTION 9

A recent study of populations of the house mouse, *Mus musculus*, on the island of Madeira resulted in the following observations:

- There are six distinct populations.
- The mice are associated with human migration and settlements.
- The populations are located in different valleys separated by steep mountains.
- Each population has a different diploid number of chromosomes

As a result of these observations, it has been suggested that speciation is taking place.

Fig. 9.1 is a schematic representation of Madeira showing the distribution of the six populations.

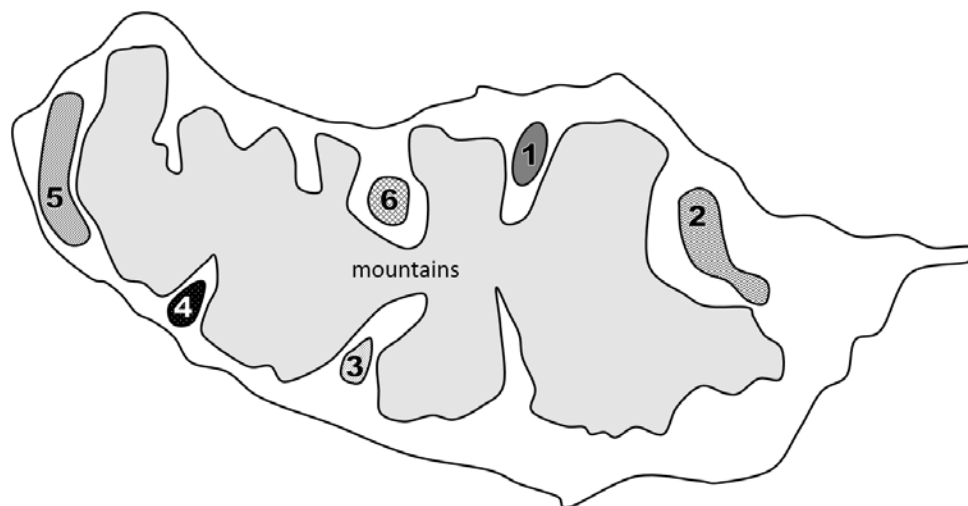


Fig. 9.1

- (a) Using the information in Fig. 9.1, state the likely isolating mechanism and the type of speciation taking place. [1]

isolating mechanism **geographical isolation**

type of speciation **allopatric speciation**

(b) 'It has been suggested that speciation is taking place.'

Explain how this process is occurring in the house mouse populations of Madeira. [5]

1. Variation exists within the ancestral population due to random mutations.
2. Due to association with human settlements, the mouse population was scattered into 6 different populations.
3. Steep mountains serves as geographical barriers for the scattered populations, hence no breeding / gene flow between the separated populations.
4. Different selection pressures in each area where each population resides
5. Mice with alleles that confer selective advantage survive to reproductive age, and pass on these alleles to offspring (viable and fertile).
6. Results in change in allele frequency / gene pool.
7. Different mutations occur independently in the 6 different populations, eventually develop different chromosome numbers.
8. Accumulation of sufficient genetic differences, hence unable to interbreed to produce viable and fertile offspring.

(c) Explain the likely outcome of individuals from two separate populations being mated in captivity. [2]

- **[Compulsory]** Due to different chromosome number / diploid number
- No homologous chromosomes, hence no pairing takes place. Meiosis cannot take place, thus no gametes will be produced by offspring, hence viable but infertile.

(d) House mouse is classified as class Mammalia, phylum Chordata, kingdom Animalia.

State one feature of the **cells** of the kingdom Animalia that distinguish them from the cells of other multicellular eukaryotes. [1]

[Comparing to the kingdom Plantae]

- Presence of centrioles/centrosomes / No cell wall / No large central vacuoles

(e) The evolutionary relationship between organisms is based on the hypothesis that the rate of mutation of DNA stays constant. The rate of mutation can be estimated by comparing the differences in amino acid sequences between species whose time of speciation is independently determined from the dating of fossils.

Explain why amino acid sequences of proteins could reveal useful evolutionary data for taxonomists. [2]

- Since DNA codes for amino acids, mutations result in altered DNA sequences, which results in differences in amino acid sequences
- Hence, large (small) difference in amino acid sequence between two species reflects distant (close) evolutionary relationship

[Total: 11]

• END OF PAPER •